

Balancing Descriptiveness and Formalization in Error Annotation: a Case of Russian Learner Corpus*

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Abstract. Recent learner corpora aimed at NLP applications often adopt highly formalized and standardized annotation schemes. While this formalization improves surface-level consistency, it frequently comes at the cost of linguistic adequacy, resulting in oversimplified error categories that fail to capture the true complexity of learner language. Analysis of the Russian Learner Corpus (RLC) reveals that such rigid schemes lead to several critical issues: artificially sharp boundaries between categories, loss of important linguistic distinctions, forced tag assignments in ambiguous cases, and, consequently, low inter-annotator agreement in linguistically rich areas. These problems ultimately degrade both the reliability of manual annotation and the quality of downstream NLP models trained on the corpus. To address these limitations, we propose a revised annotation framework that restores a better balance between formal consistency and linguistic richness. The new scheme introduces clearer yet more flexible category boundaries, reduces harmful tag overlap through explicit decision rules for difficult cases, and reorganizes error types into five linguistically motivated domains. Pilot annotation experiments demonstrate a consistent increase in inter-annotator agreement across multiple metrics. The revised framework produces more stable, linguistically adequate, and NLP-friendly annotations, making the Russian Learner Corpus more valuable both for corpus-based research and for training language models.

Keywords: Learner Corpus, Error Annotation, Grammatical Error Correction, Inter-Annotator Agreement, Second Language Acquisition

1. Introduction

Learner corpora have long served as a central resource in second language acquisition (SLA), enabling the empirical study of learner errors and interlanguage development. With the recent rise of natural language processing (NLP), their role has expanded significantly: corpora originally designed for linguistic analysis are now widely used as training and evaluation data for grammatical error correction (GEC) systems. This shift, however, exposes a fundamental mismatch between annotation schemes optimized for human interpretation and the requirements of machine learning.

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Traditional error annotation frameworks prioritize descriptive adequacy and linguistic interpretability. As a result, they often include overlapping categories, flexible tagging strategies, and rich contextual information. While such properties are valuable for linguistic research, they pose serious challenges for computational applications. In particular, inconsistent labeling, ambiguous tag boundaries, and low inter-annotator agreement negatively affect model learnability, training stability, and the reliability of evaluation metrics [3].

To address these challenges, we use the Russian Learner Corpus (RLC)¹ [28], a large resource of learner-written Russian texts. While the RLC has proven valuable for SLA research, its original annotation scheme was not designed with NLP applications. The release of the RLC-GEC² - a dataset specifically tailored for NLP objectives - on the basis of the Russian Learner Corpus [18] made explicit certain limitations. These included: overlapping tag categories, infrequent or underspecified labels, and inconsistent annotation practices hindered its effective use in machine learning pipelines.

In this paper, we present a revision of the RLC error annotation system aimed at improving both linguistic clarity and computational usability. Our aim is to eliminate systematic ambiguities in the tagset and define clear decision criteria for annotators. In contrast to the original design, which allowed multiple overlapping tags, the revised scheme emphasizes mutually exclusive categories and reproducible annotation guidelines.

The core of this work consists of (i) a detailed analysis of conflicting and co-occurring error tags in the original RLC annotation scheme, (ii) the development of a decision framework for resolving annotation ambiguities, and (iii) the introduction of revised guidelines that support consistent and machine-readable labeling.

Importantly, the primary objective of this study is not to evaluate downstream grammatical error correction systems, but to improve the consistency and reproducibility of expert error annotation. Our motivation stems from previous observations that the original RLC annotation scheme contains overlapping and underspecified categories, which lead to low inter-annotator agreement and create difficulties for automatic processing. In particular, previous work [44] showed that GEC models trained on data annotated according to the original RLC tagset achieved limited performance, suggesting that overlapping categories, underspecified labels, and inconsistent annotation practices reduce the usefulness of the resulting annotations for automatic error correction. Building on these findings, we focus on the annotation layer itself and propose a methodology for revising complex learner-error annotation schemes. The proposed approach identifies clusters of competing tags, analyzes linguistic phenomena that admit multiple plausible interpretations, and introduces explicit decision rules that delimit the scope of each category. We therefore view improved NLP usability as a motivation and expected consequence of the revision, while the empirical evaluation presented in this paper is restricted to annotation consistency and inter-annotator agreement.

The remainder of the paper is structured as follows. Section 2 introduces the Russian Learner Corpus. Section 3 describes the original annotation scheme and its limitations. Section 4 presents the revised guidelines and analyzes key sources of ambiguity across

¹ <https://www.web-corpora.net/RLC/>

² <https://github.com/Russian-Learner-Corpus/rlc-annotated>

major linguistic domains. Sections 5 and 6 report the results of annotation experiments. Section 7 concludes and outlines directions for future work.

This paper makes the following contributions:

1. **Systematic analysis of annotation inconsistencies:** We identify and categorize recurrent sources of ambiguity and inconsistency in the original RLC error annotation scheme, including overlapping tags, underspecified categories, and unstable tag combinations.
2. **Revised error taxonomy for learner corpora:** We propose a restructured annotation framework organized around five major linguistic domains, reducing redundancy and improving the separability of error categories.
3. **Decision framework for annotation disambiguation:** We introduce explicit guidelines and decision rules that resolve common conflicts between competing tags, enabling more consistent and reproducible annotation.
4. **Annotation guidelines optimized for NLP applications:** We redesign the annotation scheme to support machine-readable, non-overlapping labels suitable for training and evaluating GEC systems.
5. **Empirical evaluation via inter-annotator agreement:** We conduct pilot annotation experiments demonstrating that the revised guidelines lead to improved inter-annotator agreement across multiple error types.
6. **Resource contribution (if applicable):** We provide an updated version of the RLC annotation guidelines and release the corresponding data to support future research in SLA and NLP³.

2. Related Work

Error annotation in learner corpora has been handled in different ways depending on the intended use of the resource, resulting in a spectrum of annotation strategies that trade off linguistic descriptiveness against formal consistency and usability.

A number of large-scale learner corpora have adopted linguistically rich annotation schemes. For instance, the Cambridge Learner Corpus [25] employs a coding system in which error types are represented by a combination of error category and target word class.

Similarly, the Czech learner corpora AKCES/CZESL [38] implement a multi-level annotation framework that distinguishes between graphemic/morphemic errors, morphosyntactic errors, and higher-level phenomena such as word order and phraseology. As noted by the authors, this design represents a compromise between descriptive adequacy and practical constraints, including annotator expertise and annotation effort. The resulting system captures a wide range of linguistic phenomena, but necessarily introduces heterogeneity in category granularity.

The FALKO corpus [30] takes a different approach: it allows several possible corrections, from a minimal one (TH1) to an extended one (TH2). This framework explicitly models the non-uniqueness of “correct” forms in learner language, thereby increasing

³ <https://github.com/PolinaHitrun/revising-rlc-annotation>

descriptive richness. However, such multiplicity of interpretations further complicates the mapping between errors and labels in downstream computational tasks.

In contrast to these linguistically oriented resources, recent datasets developed for grammatical error correction (GEC) tasks tend to favor more formalized annotation schemes. For instance, in newly released datasets of L2 Russian initially created for NLP tasks, like RULEC-GEC [32,31] and ReLCo [16,15], error labels are closely linked to part-of-speech categories and grammatical features. This design enhances label specificity and aids model training, but might fail to capture linguistic interpretability of error tags [18].

Taken together, prior work reveals a fundamental tension between two design paradigms: (i) descriptive annotation frameworks, which aim to capture the full complexity of learner language but allow overlapping categories and flexible interpretation, and (ii) formalized annotation schemes, which prioritize consistency, separability of labels, and compatibility with machine learning models.

This work builds on this literature by explicitly addressing this tension. Rather than fully adopting either paradigm, we propose a revised annotation framework that aims at preserving linguistically meaningful distinctions while enforcing explicit decision rules and non-overlapping categories. In doing so, we seek to bridge the gap between annotation schemes designed for linguistic analysis and those optimized for NLP applications.

3. RLC and its Original Annotation Scheme

3.1. The Russian Learner Corpus

This study draws on the Russian Learner Corpus (RLC) [28], a large-scale collection of texts produced by learners of Russian in diverse educational contexts worldwide. The corpus includes materials collected through instructors of Russian as a second language working outside Russia, as well as contributions from the Russian Language Centre and the International Preparatory Year Programme at HSE University. At present, the RLC comprises over two million tokens (193.189 sentences), representing the written production of both L2 learners of Russian and heritage speakers—bilingual individuals for whom Russian is a home language but not the dominant one. The corpus covers speakers of 48 distinct dominant languages, reflecting a wide range of linguistic backgrounds.

The compilation of the corpus relies on long-standing institutional and personal collaborations between the RLC team and universities where Russian is taught as a foreign language. As a result, the distribution of represented language backgrounds is shaped by these collaborative networks. The RLC also incorporates RULEC, a substantial longitudinal subcorpus of L2 academic writing collected over a four-year period [42].

All participants provided informed consent, and personal identifiers were removed to ensure anonymity. The corpus is accompanied by rich metadata describing both authors and texts, including information about learners’ linguistic backgrounds and learning conditions. A substantial portion of the corpus has been manually annotated for error types and supplied with corresponding corrections.

Originally designed as a searchable resource for linguistic research, the RLC provides detailed and linguistically informed annotations that support the study of learner

language and interlanguage development. However, this design also introduces challenges when the corpus is used in computational settings, as discussed below.

3.2. The RLC-GEC Dataset

The RLC-GEC dataset [18] is a curated subset of the Russian Learner Corpus specifically adapted for grammatical error correction (GEC) tasks. It provides aligned pairs of erroneous and corrected sentences, enabling its use as training and evaluation data for supervised machine learning models.

In contrast to the original corpus interface, which was primarily designed for linguistic exploration, RLC-GEC is structured as a standalone dataset suitable for automatic processing. This transformation makes the resource accessible to the NLP community and facilitates its integration into standard GEC pipelines.

3.3. Overview of the Original Tagging System

The original annotation scheme of the RLC was inspired by learner corpora such as AKCES/CZESL and FALKO [30,37], as well as research on developmental errors in L1 Russian [41], speech errors in monolingual adult production, studies of adult heritage speakers [27], and pedagogical feedback from instructors of Russian as a foreign language [28]. The scheme aimed to prioritize flexibility, interpretability, and broad coverage of complex linguistic phenomena. The tagset distinguished five broad categories of learner errors: orthographic, morphological, syntactic, lexical, and constructional errors. In addition to these primary categories, a set of auxiliary tags can be assigned to further specify error properties. Each category captures a wide range of linguistic phenomena. For example, orthographic errors include spelling inconsistencies, hyphenation errors, and alphabet mixing; morphological errors include incorrect inflectional forms, agreement violations, and misuse of grammatical categories such as gender or number, etc.; lexical errors include inappropriate word choice; and constructional errors cover higher-level structural patterns.

The design of the tagset reflects a linguistically rich approach, that allows annotators to capture multiple aspects of a single error through the use of overlapping tags.

While the original annotation scheme was sufficient for linguistic analysis, it lacked strict guidelines for tag selection, which became critical in the context of NLP.

Annotation inconsistency introduces noise into training data, reducing model learnability and stability. Furthermore, low inter-annotator agreement undermines the reliability of evaluation benchmarks, making it difficult to compare models or reproduce results.

These challenges highlight a fundamental mismatch between descriptive annotation practices and computational requirements, motivating the need for a revised annotation framework that supports both interpretability and formal consistency.

3.4. Sources of Annotation Ambiguity and Design Goals for Revision

Our analysis identifies several recurring sources of ambiguity in the original RLC tagset:

- **Overlapping categories** applied to the same phenomenon
- **Underspecified tag boundaries**, especially between lexical and grammatical errors

- **Co-occurrence of tag pairs** without clear prioritization criteria
- **Imbalanced tag distribution**, including rare or sparsely populated categories

These issues are not independent but interact in ways that amplify annotation inconsistency. Addressing them requires not only redefining individual tags but restructuring the annotation logic as a whole.

Based on these findings, we formulate the following design goals for the revised annotation scheme:

1. **Clarity**: Define explicit boundaries between error categories;
2. **Consistency**: Reduce annotator-dependent variation through decision rules;
3. **Non-overlap**: Minimize or eliminate redundant tag combinations;
4. **Scalability**: Ensure suitability for large-scale machine learning applications;
5. **Interpretability**: Preserve linguistic relevance and explanatory power.

The next section presents the revised annotation framework developed to meet these objectives.

4. Revised Annotation Guidelines and Error Taxonomy

4.1. Overview of the Revised Framework

We propose a comprehensive revision of the RLC annotation scheme aimed at resolving systematic ambiguities and improving annotation consistency. The revised framework is also organized around five major linguistic domains: orthography, morphology, syntax, constructions, and lexis. However, in contrast to the original scheme, the revised taxonomy enforces clear boundaries between domains and prioritizes mutually interpretable and operationally consistent tag assignment. While full mutual exclusivity is not always attainable due to genuine linguistic overlap, the revised guidelines minimize redundancy through explicit decision rules and controlled use of tag combinations.

A central design principle of the new scheme is that annotation reflects the locus of the error, rather than its surface realization. This allows the system to distinguish between formally similar errors that arise from different linguistic mechanisms.

The present section does not aim to provide an exhaustive description of all tags in the revised annotation scheme. Instead, we focus on a set of representative and empirically challenging cases that exhibited the highest degree of ambiguity and annotator disagreement in the original RLC annotation. These cases are particularly informative, as they reveal systematic sources of inconsistency and motivate the design of the proposed disambiguation framework.

4.2. Decision Framework for Tag Assignment

To address inconsistencies in the original annotation scheme, we introduce a decision framework that guides annotators in resolving competing interpretations. The development of this framework is grounded in an empirical analysis of annotation data, including confusion patterns and tag co-occurrence statistics, which revealed recurrent clusters of tags that were systematically conflated in practice. These findings are further

supported by annotator feedback, which identified the same tag groups as lacking clear and operational assignment criteria.

The framework is based on three principles:

1. **Error locus prioritization:** the selected tag reflects the linguistic level at which the correction applies;
2. **Minimal sufficient explanation:** each error is assigned the smallest set of tags necessary to account for the deviation;
3. **Structured disambiguation:** in cases of overlap, predefined criteria determine tag precedence or combination.

This approach replaces the earlier practice of unconstrained multi-tagging with a controlled and reproducible mapping from error to annotation.

The error domains discussed in the subsections below were selected based on empirical evidence from the evaluation of the original RLC annotation scheme. Specifically, pilot annotation experiments revealed systematically low inter-annotator agreement (IAA) for these categories, with substantial variation in tag assignment across annotators. This indicates that the original guidelines fail to provide sufficiently explicit and operational criteria for resolving competing analyses.

Low inter-annotator agreement is corroborated by qualitative evidence from expert annotator interviews. Annotators consistently identified these domains—particularly those involving interfaces between linguistic levels (e.g., orthography vs. morphology, morphology vs. syntax, and lexis vs. aspect)—as the most difficult to annotate, citing underspecified tag boundaries, lack of prioritization rules, and frequent overlap between competing categories.

Sections 4.3, 4.4, and 4.5 focus precisely on such interfaces between different linguistic domains and, at the same time, illustrate three distinct strategies for improving the annotation scheme: reorganization of tag boundaries within a single tag group (4.3), introduction of an additional tag to increase clarity and granularity of annotation (4.4), and formalization of explicit rules for assigning tag combinations (4.5).

We therefore treat these domains as critical test cases for the revised annotation framework. Rather than attempting to exhaustively document the full tagset, we focus on a set of high-impact ambiguity zones that account for a disproportionate share of annotation inconsistency. These cases serve a dual purpose: (i) to diagnose systematic weaknesses in the original scheme, and (ii) to demonstrate how the revised guidelines introduce explicit decision criteria, reduce overlap, and improve reproducibility.

The effectiveness of these revisions is evaluated quantitatively in Section 5, where we report inter-annotator agreement scores before and after the application of the new guidelines. In particular, the domains discussed below correspond to those exhibiting the largest gains in IAA, providing empirical support for the proposed restructuring of the annotation scheme.

4.3. Orthographic and Morphological Interface

Spelling Errors. Spelling errors are among the most frequent in learner corpora, and a central challenge in orthographic annotation lies in distinguishing between errors that

arise from different mechanisms [3,23,24,11]. In learner data, deviations from the target form may reflect phonological encoding, visual confusion, incomplete orthographic knowledge, or purely mechanical slips. Treating all such cases as “spelling errors” obscures important distinctions and reduces annotation consistency.

In the original annotation scheme of the Russian Learner Corpus, orthographic errors were partially subdivided into more specific tags, such as those covering hyphenation (*Hyphen*) or merged vs. separate spelling (*Space*). However, these categories remained relatively infrequent, while a substantial proportion of orthographic deviations were assigned to the general *Ortho* category, which did not differentiate between underlying error mechanisms and thus failed to capture the internal heterogeneity of this error type. The IAA score for *Ortho* tag equalled 0.163 (Cohen’s Kappa coefficient [4]).

The revised annotation scheme therefore differentiates four core subtypes of spelling deviations: **Ortho** captures errors motivated by phonological representation, i.e., cases where the learner encodes the word “as heard.” These include substitutions based on phonetic similarity or the neutralization of contrasts in weak positions:

- (1) a. *сагласен
 *saglasen
 [saglasen]
 b. согласен
 soglasen
 [saglasen]
 ‘agree’

In (1), due to phonetic reduction in pre-stressed syllables the pronunciation of the word with the original and the corrected spelling is identical. **Graph** is used for visually motivated substitutions, where the error is driven by graphemic similarity rather than sound. This includes both intra-script confusions (e.g., *u/ü*, *u/ul*) and cross-script interference (e.g., Cyrillic *з* [g] for Latin *r* [r]):

- (2) a. *реагировала
 *reagirovala
 [geagirovala]
 b. реагировала
 reagirovala
 [reagirovala]
 ‘react’

Misspell applies to complex, non-decomposable deviations, where the erroneous form cannot be reduced to a sequence of independent single-character errors. These cases often reflect unstable or incomplete lexical representations rather than a single identifiable process.

- (3) a. *менеию
*meneiû
b. мнению
mneniû
‘opinion’

Typo is reserved for performance errors, such as accidental keystrokes, where the deviation can not be explained by linguistic competence:

- (4) a. *сышал
*syšal
b. слышал
slyšal
‘heard’

This distinction allows to differentiate between phonological encoding, orthographic retrieval, and mistakes that cannot be classified as potential gaps in linguistic competence, such as failures in motor execution. By aligning annotation categories with these underlying mechanisms, the revised scheme improves both interpretability and consistency.

Spelling vs. Inflectional Errors. A major source of annotation ambiguity in learner corpora for morphologically rich languages arises at the interface between orthography and morphology, particularly in word endings. In Russian, unstressed vowels undergo phonological reduction, making it difficult to distinguish whether a learner error reflects: (i) a phonetic spelling strategy (“write as you hear” that was mentioned in the previous subsection), or (ii) a failure to select the correct inflectional form. This ambiguity is especially pronounced for L2 learners, who must simultaneously acquire phonological reduction patterns and morphological paradigms. The original tagset used tag Infl for errors in inflections, with the IAA score as low as -0.0528 (Cohen’s Kappa coefficient [4]).

To ensure consistent annotation, the revised guidelines introduce the following operational criteria:

- An error is classified as Ortho if the incorrect form can be interpreted as a phonologically plausible representation of the target word. In such cases, the learner preserves the underlying structure but misrepresents reduced vowels or neutralized contrasts:

- (5) a. *улучшет
*ulučšet
b. улучшит
ulučšit
‘improve’

- An error is classified as *Infl* if it reflects an incorrect grammatical choice, i.e., the selected ending does not belong to the paradigm required by the syntactic context:

- (6) a. *говоре~~м~~
 *govore~~m~~
 b. говор~~им~~
 govor~~im~~
 ‘speak’

A key diagnostic is the recoverability of the intended form:

- If the intended grammatical form is identifiable and the deviation is phonologically motivated, then it is classified as *Ortho*.
- Otherwise, it is classified as *Infl*.

This distinction allows annotators to systematically separate form encoding errors from grammatical selection errors, which is essential for both linguistic analysis and downstream NLP tasks.

4.4. Morphological vs Syntactic Structure

Errors in Passives and Reflexive Forms. The distinction between reflexive marking and passive voice formation presents a well-known challenge in the analysis of Russian verbal morphology [10,12,34,35,20,5]. Both phenomena involve argument structure alternations and often share similar surface markers, most notably the reflexive suffix *-ся*, which leads to systematic confusion in annotation.

In the original RLC scheme, this domain was covered with the tags *Refl* (errors in reflexive forms) and *Passive* (errors in passive constructions). These tags were applied inconsistently because they conflated both morphological errors (7) (incorrect formation of reflexive or passive forms) and syntactic errors (8) (incorrect choice of voice or errors in argument structure)

- (7) a. * Я **интересую** языком
 * Â **interesuû** âzykom
 b. Я **интересуюсь** языком
 Â **interesuûs'** âzykom
 ‘I am interested in languages’
- (8) a. * Им **создавались** множество препятствий
 * Im (INSTR) **sozdavalis'** množestvo prepâtstvîj
 b. Он **создавал** множество препятствий
 On (NOM) **sozdaval** množestvo prepâtstvîj
 ‘He created many obstacles’

This conflation reflects a deeper linguistic issue: in Russian, voice-related distinctions are not confined to morphology alone but display a complex distribution across both morphological form and syntactic argument realization [10,17,20]. The IAA score for the *Refl* tag in the original annotation was only 0.167.

To resolve this ambiguity, the revised scheme adopts a level-based separation: *Refl* is reserved for morphological errors in reflexive marking, such as incorrect presence or absence of the reflexive suffix (9). *Passive* is reserved for morphological errors in passive formation, such as incorrect use of participial forms or auxiliary constructions (10). Errors involving argument structure or voice alternation at the clause level are reassigned to the new syntactic tag *Voice* (11).

- (9) a. * Наша планета **изменяет себя**
 * Naša planeta **izmenâet sebâ**
- b. Наша планета **меняется**
 Naša planeta **menâetsâ**
 ‘Our planet changes’
- (10) a. * В целях безопасности **ввелось** карантин
 * V celâh bezopasnosti **vvelos’** karantin
- b. В целях безопасности **был введен** карантин
 V celâh bezopasnosti **byl vveden** karantin
 ‘For safety reasons quarantine was introduced’
- (11) a. * **Эти песни говорят** о разных проявлениях войны
 * Èti pesni govorât o raznyh proâvleniâh vojny
- b. **В этих песнях говорится** о разных проявлениях войны
 V ètih pesnâh govoritsâ o raznyh proâvleniâh vojny
 ‘In these songs it is said that there are different manifestations of war’

This restructuring aligns annotation with linguistic theory by distinguishing between form-level errors (morphology) and structure-level errors (syntax). As a result, the revised scheme eliminates a major source of overlap and improves category coherence.

4.5. Aspect vs Lexical Errors in Prefixed Verbs

Another domain that causes persistent challenges in both Second Language Acquisition (SLA) and learner corpus annotation concerns the treatment of verbal aspect. A substantial body of research has shown that aspectual distinctions are difficult for L2 learners, as they require the integration of temporal interpretation, event structure, and grammatical marking [1,33]. These difficulties are particularly pronounced in Slavic languages

such as Russian, where aspect is deeply embedded in the verbal system and interacts systematically with derivational morphology.

Empirical studies on L2 Russian demonstrate that learners struggle not only with selecting the appropriate aspectual form, but also with interpreting and producing aspectually marked verbs in context [36,39,8]. Crucially, these difficulties persist even at advanced proficiency levels and are attested in both L2 learners and heritage speakers. This suggests that the source of the problem lies not merely in surface complexity, but in the structural properties of the Russian aspectual system.

An important factor contributing to this complexity is the non-trivial interaction between prefixes and verbal stems in Russian, as well as the existence of non-prefixal aspectual derivational mechanisms such as suffixation, root alternations, and suppletivism [21,22,2,43,29,9]. Many verbal prefixes are polysemous and may substantially affect the verb semantics and its argument structure [6,26,14,40].

This theoretical indeterminacy has direct consequences for error annotation in learner corpora. Evidence from the Russian Learner Corpus (RLC) shows that learner errors involving aspectually marked verbs frequently resist unambiguous classification. Annotators often hesitate between *Lex* and *Asp* tags for formally similar cases, reflecting not only inconsistency in annotation practice but also genuine ambiguity inherent in the Russian aspectual system. In many instances, it is unclear whether a non-target form results from incorrect lexical selection (wrong prefix meaning), incorrect aspectual marking, or an interaction of both.

To address this issue, the revised annotation guidelines adopt the following disambiguation strategy:

- The *Asp* tag is used when the error primarily involves the choice of aspectual value (perfective vs. imperfective) in a context where the lexical meaning of the verb remains appropriate (12)⁴.
- The *Lex* tag is used when the error involves incorrect prefix selection that alters the lexical or phasal meaning of the verb, while aspectual marking itself is appropriate (13).
- In cases where both dimensions are simultaneously violated and cannot be cleanly separated, the combined *Lex + Asp* annotation is permitted (14).

- (12) a. * Он всегда улы**н**ётся
 * Он vsegda ulybnëtsâ
 b. Он всегда улы**б**ается
 On vsegda ulybaetsâ
 ‘He always smiles’

⁴ Judging aspectual errors is a separate methodological issue. The examples given in this section were rated as erroneous by multiple annotators given their specific contexts in discourse.

- (13) a. * Будет **смотреть тебя**, когда ты нездорова
 * Budet **smotret' (IPF) tebâ**, kogda ty nezdorova
- b. Будет **присматривать за тобой**, когда ты нездорова
 Budet **prismatrvat' (IPF) za toboj**, kogda ty nezdorova
 '(S)he will look after you when you are unwell'
- (14) a. * До 1989-ого года, стена **отделила** не только страны и идеологии,
 * Do 1989-ogo goda, stena **otdelila** ne tol'ko strany i ideologii,
 но и семьи
 no i sem'i
- b. До 1989-ого года стена **разделяла** не только страны и идеологии,
 Do 1989-ogo goda stena **razdelâla** ne tol'ko strany i ideologii,
 но и семьи
 no i sem'i
- 'Up to 1989, the wall had been separating not only ideologies but also families'

This approach acknowledges the gradient nature of prefixal semantics in Russian while maintaining operational clarity. Rather than forcing an artificial binary distinction, the guidelines allow annotators to capture both dimensions of the error when necessary, while still prioritizing consistent and interpretable labeling.

Overall, this case illustrates a broader point: annotation ambiguity is not always a flaw of the scheme, but may reflect genuine indeterminacy in the linguistic system itself. The goal of the revised framework is therefore not to eliminate all ambiguity, but to make it explicit, controlled, and reproducible in annotation practice.

5. Inter-Annotator Agreement

To evaluate whether the revised annotation guidelines improved inter-annotator agreement, we conducted two small-scale pilot experiments. All the data and code for running experiments are publicly available on GitHub. The first one was focused on orthographic errors and the second one was focused on all errors, and both were limited in size, but they already provide an initial indication of how the new system may affect consistency. In the future, we aim to expand these experiments with larger datasets and more annotators in order to obtain statistically robust results.

In the first experiment, annotators concentrated on three tags: Ortho, Infl, and Misspell. Two annotators used the old guidelines and two used the new ones. They were given sentences from RLC tagged with these three target tags. Agreement was measured using both Fleiss' Kappa (1)

$$\begin{aligned}
\kappa_F &= \frac{\bar{P} - \bar{P}_e}{1 - \bar{P}_e} \\
\text{where } \bar{P} &= \frac{1}{N} \sum_{i=1}^N p_i, \quad p_i = \frac{1}{n(n-1)} \sum_{j=1}^k n_{ij}(n_{ij} - 1) \\
\text{and } \bar{P}_e &= \sum_{j=1}^k p_j^2, \quad p_j = \frac{1}{Nn} \sum_{i=1}^N n_{ij}
\end{aligned} \tag{1}$$

N is the total number of errors, n is the number of annotators, and k is the number of possible tags

and Cohen's Kappa (2)

$$\begin{aligned}
\kappa_C &= \frac{P_o - P_e}{1 - P_e} \\
\text{where } P_o &= \frac{1}{N} \sum_{i=1}^N (a_i = b_i) \\
\text{and } P_e &= \sum_{i=1}^k p_a(i) \cdot p_b(i)
\end{aligned} \tag{2}$$

a and b are two annotators and N is amount of errors

as there were only two annotators [7,4]. Importantly, the analysis did not take into account cases where an error was identified by some annotators but not by others. Secondary tags were ignored in this experiment. Under the old guidelines, Fleiss' Kappa was 0.4955 and Cohen's Kappa was 0.4965. Under the new guidelines, Fleiss' Kappa rose to 0.5411 and Cohen's Kappa to 0.5455. Standard errors, however, were large, and a z-test revealed no statistical significance neither for Fleiss' Kappa, nor for Cohen's. Thus, while agreement improved, the effect cannot be confirmed with confidence due to the limited sample size and high standard error.

The second experiment approached agreement from a different angle. We selected three learner texts and manually pre-identified all error slots. Annotators were asked only to assign tags to these positions. This design focused exclusively on label choice. As in the first experiment, all cases where at least one annotator judged that no error was present were removed. Each text was annotated by three annotators following the old guidelines and three following the new. Agreement for individual texts was measured using Fleiss' Kappa (1), pairwise Jaccard similarity (3) [13], and Krippendorff's Alpha (4) [19], transformed for multilabel annotation. For the concatenated dataset, only two annotators overlapped across all three texts, so Cohen's Kappa (2) was calculated instead of Fleiss'. However, Fleiss' and Cohen's Kappa do not support multilabel annotation.

$$\bar{J} = \frac{\sum_{i,j} \sum_{s \in S} J(A_i(s), A_j(s))}{C_n^2 |S|} \quad (3)$$

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

$A_i(s)$ and $A_j(s)$ are the sets of tags assigned to error s by annotators i and j , S is the set of all errors, and C_n^k is the number of unique annotator pairs

$$\alpha = 1 - \frac{D_o}{D_e} \quad (4)$$

where $D_o = \frac{1}{N} \sum_s \frac{1}{k_s - 1} \sum_c (c_{si} = c_{sj})$

and $D_e = 1 - \sum_c P(c)^2$

k_s is the number of annotations for an error s , c_{si} and c_{sj} are the tags for error s by annotators i and j , and $P(c)$ is the proportion of annotations in category c

The results show a consistent upward trend across nearly all metrics. All the results are shown in Table 1 and Figure 1. For the first text, Fleiss' Kappa increased from 0.2283 to 0.3520, with Jaccard similarity rising from 0.4457 to 0.5286. Krippendorff's alpha showed a slight decrease for the first text from 0.2786 to 0.2579. Other two texts exhibited the same pattern: Fleiss' Kappa rose from 0.1808 to 0.3733 for text 2 and from 0.2105 to 0.3511 for text 3, accompanied by comparable increases in Jaccard similarity and Krippendorff's Alpha values. The concatenated dataset also reflected this trend: Cohen's Kappa increased from 0.2131 to 0.4295, with Jaccard similarity rising from 0.3392 to 0.5412. Overall, agreement increased across all metrics, with the sole exception of Krippendorff's Alpha for text 1.

Table 1. Inter-annotator agreement metrics for old and new guidelines

Text ID	Fleiss' Kappa		Jaccard pairwise		Krippendorff's Alpha	
	old	new	old	new	old	new
1	0.2283	0.3520	0.4457	0.5286	0.2786	0.2579
2	0.1808	0.3733	0.2754	0.5495	0.0961	0.2583
3	0.2105	0.3511	0.4372	0.4846	0.1384	0.2789
All texts	0.2131	0.4295	0.3392	0.5412	0.1974	0.2299

To further assess the robustness of the observed improvements, we conducted a non-parametric permutation test (2,000 iterations) and bootstrap resampling to estimate 95% confidence intervals on the concatenated dataset. Both analyses supported the observed increase in inter-annotator agreement under the revised guidelines. The improvements

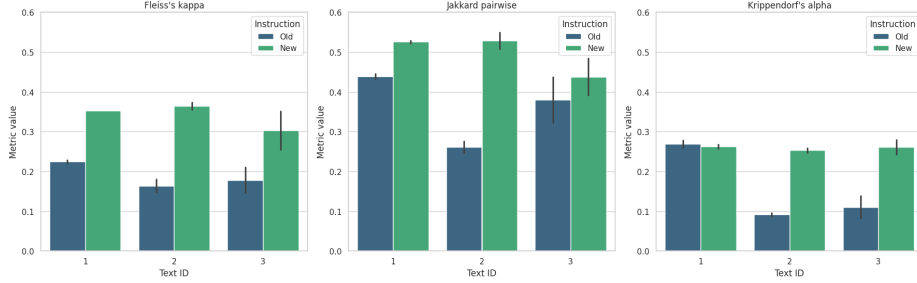


Fig. 1. Inter-annotator agreement metrics for old and new guidelines

in Fleiss' kappa and pairwise Jaccard similarity were consistently positive and unlikely to arise from random variation alone. While these results do not eliminate the limitations imposed by the pilot scale of the study, they provide additional evidence that the observed trend is systematic rather than accidental.

We also note that the slight decrease in Krippendorff's alpha for Text 1 should be interpreted with caution. In the multilabel setting used here, alpha was computed over a highly imbalanced tag inventory with a substantial long-tail distribution of rare labels. As a result, the metric is particularly sensitive to disagreement on infrequent categories. By contrast, Fleiss' kappa and instance-level Jaccard similarity are less affected by label sparsity and therefore provide a more stable estimate of overall agreement in this setting.

To conclude, both experiments point in the same direction: inter-annotator agreement improves when annotators use the revised guidelines. The consistency of the trend suggests that the updated annotation scheme is more internally coherent and easier to apply. Future work will scale up the evaluation by involving more annotators.

6. Inter-Model Agreement Experiment

To complement the human annotation study described in the preceding section, we conducted a parallel inter-model agreement experiment in which large language models (LLMs) served as annotators. Unlike the human label-selection task (task 2 from the experiments), which controlled for error detection by providing pre-annotated error positions, the LLM experiment presented models with raw learner texts and required them to perform both error detection and error classification simultaneously – more closely replicating an unconstrained annotation scenario.

Three state-of-the-art models served as annotators: DeepSeek-V3, Gemini 2.5 Flash and Claude Opus 4. All models received a russian prompt instructing them to identify all errors in the text, assign tags according to the supplied guidelines, and return their output as a CSV table with columns for the containing sentence, the erroneous span, and the assigned tag(s). When multiple tags applied to a single span, models were instructed to combine them with a + separator; when a sentence contained multiple errors, each was listed on a separate row. Beyond the prompt, each model received either the original or the revised annotation guidelines together with the target text.

The experiment covered the same three learner texts used in the human study. Each unique (text $\times$ model $\times$ guideline version) combination was run three times in order to capture intra-model variability alongside inter-model agreement, yielding 54 annotation sets in total.

Because models performed full annotation rather than label selection over a fixed set of spans, the resulting outputs could not be directly compared: models frequently identified the same underlying error but with different span boundaries or surrounding context. To address this, all model outputs were manually reviewed and reconciled into a unified dataset, aligning spans that referred to the same error before computing agreement. Agreement was then measured over this unified set of error positions using Fleiss’ κ and Krippendorff’s α , applied consistently across individual texts and the concatenated dataset. Mean pairwise Jaccard similarity was not computed, as it requires a shared fixed set of spans and is therefore for our task of comparing two instructions is not relevant.

The overall pattern over the three texts is broadly consistent with the findings of the human annotation experiment (see Table 2); when all texts were considered jointly, both agreement metrics increased under the revised guidelines. These results, however, didn’t prove statistically significant on applying the same non-parametric permutation test and bootstrap resampling procedure as in the human annotation experiment (Fleiss’ κ increased by 0.0212 ($p = 0.128$) and Krippendorff’s α by 0.0305 ($p = 0.197$)). So given the small scale of the experiment, these results should be interpreted as preliminary evidence of a positive trend rather than conclusive confirmation of improved agreement.

Table 2. Inter-model agreement metrics for old and new guidelines

Guidelines	Fleiss’ κ	Krippendorff’s α
Old annotation	0.3997	0.4037
New annotation	0.4210	0.4380

Intra-model consistency – the agreement of a single model with itself across three independent runs – is uniformly higher than inter-model agreement, confirming that disagreement arises primarily between models rather than within them. Under the revised guidelines, all three models become more self-consistent: DeepSeek (Jaccard: 0.1717 $\rightarrow$ 0.2236), Claude (0.2016 $\rightarrow$ 0.2603), and Gemini (0.2644 $\rightarrow$ 0.2938).

Compared to the human experiment, LLM κ -values under both guideline versions are generally higher than the corresponding human figures. Unlike the human study, Text 2 shows reduced agreement among models and merits further investigation. The next step would be a detailed analysis of the annotation disagreements, identifying which specific error tags drive the greatest inconsistency across models and whether these overlap with the tags that proved most problematic for human annotators.

7. Conclusion

We presented a revision of the error annotation scheme for the Russian Learner Corpus (RLC). The proposed framework addresses recurrent sources of annotation inconsistency, including overlapping categories, underspecified tag definitions, and the absence of explicit decision criteria. To this end, we developed a methodology for identifying clusters of competing tags and introducing disambiguation rules that clarify their scope. The resulting scheme establishes clearer category boundaries and promotes more consistent annotation practices while preserving linguistically meaningful distinctions.

We showed that major sources of disagreement concentrate in linguistically complex domains, including the orthography–morphology interface, the morphology–syntax boundary, and the lexicon–grammar continuum. By analyzing clusters of competing tags and formalizing decision rules for their assignment, the revised scheme reduces ambiguity and improves the reproducibility of expert annotation.

Pilot experiments demonstrated consistent improvements in inter-annotator agreement across multiple metrics, suggesting increased internal coherence and usability of the revised guidelines. Although the empirical evaluation remains limited in scale, the observed improvements were consistent across agreement measures and provide evidence that the proposed framework is easier to apply in practice.

The primary contribution of this work lies in the annotation layer itself. Rather than evaluating downstream GEC systems, we focused on developing a principled methodology for revising learner-error annotation schemes and supporting expert annotators in resolving linguistically motivated annotation conflicts. These results are relevant for NLP applications, as annotation consistency is a prerequisite for creating reliable training and evaluation data.

At the same time, some ambiguity reflects genuine properties of the linguistic system and cannot be fully eliminated. Future work will focus on scaling annotation, extending the revised scheme to the full corpus, and evaluating the impact of the revised annotation framework on downstream GEC performance.

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